

MEMORANDUM

10.02.2026

PART XIII – MEDICAL RULES

Rules amendments applying on **10.02.2026**

Chapter III PROTECTION AND PROMOTION OF THE RIDER'S HEALTH

§6 Ban of monoxide carbon inhalations outside medical facilities

13.3.066 Introduction

The use of systems or equipment allowing the inhalation of carbon monoxide is prohibited both in- and out-of-competition (category M1.4 of the Prohibited Methods List).

~~Carbon monoxide (CO) is a poisoning colorless and odorless gas that binds to haemoglobin (Hb) with a very high affinity, approximately 200-fold higher than that of oxygen. Normal carboxyhaemoglobin concentrations do not exceed 2% in non-smokers. When inhaling CO, most of the gas diffuses within the first seconds from the alveoli to the arterial blood, binds to Hb, decreases the oxygen carrying capacity and induces hypoxia.~~

~~Despite being associated with toxic effects, CO is commonly used in sports medicine to measure the total Hb mass and blood volume, especially to examine the effects of endurance training and altitude exposure on the oxygen carrying capacity.~~

~~However, CO is a toxic gas at high doses. Moreover, its repeated inhalation can result in serious acute and chronic health problems. Symptoms of CO toxicity from acute exposure, include headache, lethargy, nausea, dizziness, and confusion. In addition, given the wide inter-individual variability in responses to CO, there is a risk of clinical signs of intolerance.~~

~~In light of the foregoing, in order to protect the health and physical integrity of the athletes, CO inhalations are prohibited, except in medical facilities and under the supervision of a qualified healthcare professional experienced in the handling and use of this gas for medical purposes.~~

~~13.3.067~~ Scope of the regulation

~~The following provisions apply to all licence holders and/or entities subject to UCI Regulations, both within and outside the framework of cycling events. These provisions also apply autonomously and independently of any other regulations.~~

~~When reviewing the facts and the law of a given case, courts, arbitral hearing panels and other adjudicating bodies should be aware of and respect the strictly medical purpose of this rule, which is to protect the athletes' health, as defined in this article.~~

~~The ban on CO inhalation as set out in this section came into force on 10 February 2025.~~

13.3.0678 Authorised use of CO inhalation

The inhalation of carbon monoxide is not prohibited when it is administered as part of a diagnostic procedure, especially for determining the total haemoglobin mass and blood volume.

~~A single inhalation of CO is authorised to determine total Hb mass and blood volume, in accordance with a scientifically and medically validated protocol.~~

CO inhalation must therefore be carried out in medical facilities (medical service, medical office, or a location equipped with tools and materials to handle any medical emergencies) and under the supervision of a qualified healthcare professional experienced in the handling and use of this gas for medical purposes.

~~A CO inhalation to~~The measure total Hb mass should only be carried out in duplicate~~once~~. A second CO inhalation, and therefore a second measure of total Hb mass (and then CO inhalation), and therefore a second measure of total Hb mass, may however be carried out two weeks after the initial measurement.

For riders of UCI WorldTeams and UCI ProTeams, any single CO inhalation for the purpose of determining total Hb mass must be recorded in the medical file established for each athlete, in compliance with Articles 13.3.020 to 13.3.026 of the UCI Medical Rules.

All other CO inhalation protocols are prohibited.

~~13.3.069~~ Infringements of the ban of CO inhalation

~~The following are considered to be infringements of the ban on CO inhalation laid down in this section:~~

- ~~a) the possession, outside any medical facility, of commercially available CO re-breathing systems connected to oxygen and CO cylinders.~~
- ~~b) the use of such systems outside any medical facility and/or without the supervision of a qualified healthcare professional experienced in the handling and use of this gas for medical purposes.~~

~~For the purposes of this provision, "possession" may be understood as the holding of such a device by a team, a rider, a team member, including a medical staff, or any other individual possessing such equipment on behalf of riders.~~

~~13.3.070~~ Sanctions

~~The UCI Disciplinary Commission shall be competent to sanction any violation of the ban on CO inhalation.~~

~~Any person or entity subject to Title XII of the UCI Regulations who possesses or uses, in accordance with the provisions of article 13.3.069 of the present regulations, a CO re-breathing system connected to oxygen and CO cylinders, or on whose behalf such a system is possessed or used, shall be liable to one or more of the disciplinary measures provided for in articles 12.2.002 et seq.~~

~~The proven use of repeated inhalations of CO, and/or any inhalation not intended for the determination of total Hb mass and blood volume, will be considered an aggravating factor in determining the type and extent of the disciplinary measures in accordance with article 12.2.003.~~